

AI Fitness App Knowledge Base

Training, nutrition, progression, supplements, body-fat estimation, coaching memory, and safety guardrails for an adaptive AI fitness coach.

Purpose: This document is designed to be uploaded into an app knowledge base. It uses explicit rules, substitution maps, decision logic, and response patterns so the AI can generate safer, more useful, and more personalized fitness coaching. It is not medical advice.

Recommended app usage: chunk by module, preserve headings, and let the AI retrieve the relevant module based on user goal, equipment, pain, nutrition constraints, supplement questions, or progress logs.

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1. Core Operating Principles

The app should not behave like a static template generator. It should behave like an adaptive coach that updates training, nutrition, and recommendations based on user goals, equipment, limitations, logs, recovery, preferences, and adherence patterns.

Universal principles

- Specificity drives adaptation: train the outcome the user wants to improve.
- Progressive overload is required over time, but overload can come from load, reps, sets, range of motion, tempo, density, skill, or consistency.
- Volume must match recovery. More work is not better if performance, sleep, joints, or motivation decline.
- The split is only the weekly organization. The drivers are goal-specific volume, intensity, frequency, effort, recovery, and adherence.
- Pain should trigger modification, not force. The app should never encourage users to push through sharp or worsening pain.
- The best plan is the one the user can repeat consistently while progressing.

AI response posture

- Use confidence language: likely, approximately, based on current information, visual estimate, or coaching estimate.
- Avoid false precision. Do not claim exact body fat, exact calorie burn, guaranteed outcomes, or medical certainty.
- When risk is low, give clear action. When risk is high, encourage professional help and offer safe alternatives.
- Use the user's stated goal first, but adjust based on body-fat estimate, experience, equipment, recovery, and limitations.

2. Safety and Medical Guardrails

Hard safety rules

- Never diagnose injuries, diseases, hormonal issues, eating disorders, or medical conditions.
- Never override a clinician, physical therapist, registered dietitian, or medical restriction.
- Never prescribe extreme deficits, dehydration, laxatives, diuretics, steroid cycles, SARMs, prohormones, or fat burners.
- Never encourage training through sharp pain, numbness, tingling, swelling, dizziness, chest pain, or shortness of breath beyond normal exertion.
- Never frame the user's body as bad, disgusting, lazy, or broken.
- Do not provide aggressive weight-loss guidance to minors, pregnant users, or users describing eating-disorder behaviors. Redirect to professional support.

Escalation triggers

- Chest pain, fainting, severe dizziness, or unusual shortness of breath during exercise.
- Sudden pop, swelling, acute injury, or inability to bear weight.
- Numbness, radiating pain, loss of function, or neurological symptoms.
- Rapid unexplained weight loss, disordered eating, purging, laxative use, or obsession with extreme leanness.

- Pregnancy, post-surgery training, cardiac history, uncontrolled hypertension, diabetes medication changes, or kidney disease with supplement questions.

3. Training Goal Framework

Goal	Primary Driver	Best Split Patterns	Key Metrics
Fat loss	Calorie deficit, protein, lifting, steps, sustainable cardio	3-day full body; 4-day upper/lower; 2 lifting + 3 cardio	Weight trend, waist, photos, steps, workout completion
Muscle gain	Hypertrophy volume, progressive overload, protein, recovery	3-day full body; 4-day upper/lower; 5-day hybrid; 6-day PPL	Load/reps, measurements, photos, bodyweight trend
Aesthetics	Hypertrophy plus fat-loss control and weak-point bias	Upper/lower aesthetic; PPL; physique split	Photos, waist, shoulders, chest, glutes, arms, definition
Strength	Heavy specific lifts, skill practice, lower reps, longer rest	3-day full body; 4-day upper/lower; powerlifting split	Estimated 1RM, top sets, technique, volume PRs
Endurance	Aerobic base plus purposeful intensity	3-6 cardio days; polarized/pyramidal; hybrid strength	Pace, distance, duration, HR zones, RPE
Athletic	Strength, power, mobility, conditioning, skill transfer	Hybrid strength + conditioning; performance blocks	Sprints, jumps, agility, strength, conditioning tests

Goal-specific training anchors

- Strength: prioritize heavy compound patterns early in the session. Use lower reps, longer rest, and avoid frequent failure on technical lifts.
- Hypertrophy: prioritize weekly hard sets per muscle, controlled execution, full useful range of motion, and sets close to failure on appropriate movements.
- Fat loss: maintain or build muscle through lifting while using steps, cardio, and nutrition to support the deficit.
- Endurance: most work should be easy enough to recover from, with 1-2 hard sessions weekly depending on level.
- Aesthetics: train for hypertrophy while biasing visual muscles and managing body fat through nutrition and cardio.

4. Limitation and Pain Handling

```
type LimitationSeverity =
  | 'none'
  | 'mild_discomfort'
  | 'recurring_pain'
  | 'acute_pain'
  | 'medical_restriction'
  | 'post_surgery_or_rehab';
```

Pain response logic

User report	AI action	Example response
Mild discomfort	Reduce load, range, tempo, or swap variation. Avoid provoking pain.	We can switch to a more joint-friendly option and keep the movement pain-free.

User report	AI action	Example response
Recurring pain	Flag limitation, swap exercise, suggest professional assessment if persistent.	Since this keeps coming up, we should stop using that movement for now and replace it.
Sharp or acute pain	Stop aggravating exercise. Do not diagnose. Recommend medical evaluation if persistent or serious.	Sharp pain is not something to train through. Stop that movement today.
Post-surgery or rehab	Ask what clinician cleared. Use only approved movements.	I can build around your clearance, but I should not override your rehab plan.
Medical restriction	Treat as hard constraint.	We will keep the plan inside the limits you were given.

Joint-specific modification examples

- Knee-sensitive squat: use box squat, goblet squat to comfortable depth, low step-up, leg press partial range, reverse lunge, or hip-dominant work.
- Back-sensitive hinge: use hip thrust, glute bridge, cable pull-through, chest-supported row, elevated trap-bar deadlift, or reduce spinal loading.
- Shoulder-sensitive pressing: use neutral-grip dumbbell press, landmine press, machine press, cable press, push-up handles, or incline press.
- Wrist-sensitive pressing: use push-up handles, neutral grips, dumbbells, machines, or forearm-based plank variations.
- Ankle-sensitive lower body: reduce deep dorsiflexion demands, elevate heel if tolerated, use leg press, hip hinge, or step-ups with controlled range.

5. Exercise Substitution System

The substitution engine should replace movement patterns, not random exercises. A squat should be replaced with another squat/knee-dominant pattern unless pain or equipment requires a broader modification.

```
type MovementPattern =
  | 'squat'
  | 'hinge'
  | 'horizontal_push'
  | 'vertical_push'
  | 'horizontal_pull'
  | 'vertical_pull'
  | 'lunge_single_leg'
  | 'knee_flexion'
  | 'hip_extension'
  | 'calf_raise'
  | 'core_anti_extension'
  | 'core_anti_rotation'
  | 'loaded_carry'
  | 'conditioning';
```

Substitution map

Pattern	Full gym	Dumbbells only	Home/bodyweight	Limitation-friendly
Squat	Back squat, front squat, hack squat, leg press, belt squat, Smith squat	Goblet squat, DB front squat, split squat, step-up, reverse lunge	Bodyweight squat, chair box squat, split squat, wall sit	Box squat, controlled goblet squat, low step-up, partial-range leg press

Pattern	Full gym	Dumbbells only	Home/bodyweight	Limitation-friendly
Hinge	Deadlift, RDL, trap bar deadlift, hip thrust, cable pull-through, back extension	DB RDL, single-leg DB RDL, DB hip thrust	Glute bridge, single-leg bridge, hamstring walkout, backpack RDL	Hip thrust, glute bridge, cable pull-through, elevated trap bar
Horizontal push	Bench press, DB bench, machine chest press, Smith press, cable press	DB bench, DB floor press, DB incline press	Push-up, incline push-up, tempo push-up, band press	Neutral-grip DB press, machine press, floor press, cable press
Vertical push	Overhead press, machine shoulder press, landmine press, DB press	Seated DB press, half-kneeling DB press, Arnold press	Pike push-up, band overhead press, lateral raise	Landmine press, incline press, scaption raise, lateral raise
Horizontal pull	Cable row, chest-supported row, machine row, T-bar row	One-arm DB row, chest-supported DB row, rear-delt row	Band row, inverted row, backpack row	Chest-supported row, cable row, machine row
Vertical pull	Pull-up, assisted pull-up, lat pulldown, cable pullover	DB pullover, one-arm row, rear-delt row	Band pulldown, inverted row, negative pull-up	Neutral-grip pulldown, assisted pull-up, cable pullover
Single leg	Bulgarian split squat, step-up, reverse lunge, walking lunge, single-leg press	DB split squat, DB reverse lunge, DB step-up	Split squat, assisted lunge, step-up	Reverse lunge, low step-up, assisted split squat
Core anti-extension	Ab wheel, cable rollout, stability ball rollout, plank	Plank, dead bug, hollow hold	Dead bug, plank, hollow hold, body saw	Dead bug, short plank, bird dog
Core anti-rotation	Pallof press, cable hold, suitcase carry	Suitcase carry, one-arm carry	Band Pallof, bird dog, side plank	Bird dog, Pallof hold, short suitcase carry

Exercise swap rules

- Preserve the target muscle and movement pattern when possible.
- Preserve relative difficulty: do not replace a heavy squat with an easy bodyweight movement unless equipment or pain requires it.
- When load is limited, increase difficulty with unilateral work, tempo, pauses, higher reps, shortened rest, or mechanical drop sets.
- When pain is present, reduce range and load before adding complexity.
- When the user dislikes an exercise, swap within the same pattern first.

6. Equipment Profiles

```

type EquipmentProfile =
  | 'full_gym'
  | 'dumbbells_only'
  | 'home_basic'
  | 'bodyweight_only'
  | 'bands_only'
  | 'hotel_gym'
  | 'barbell_home_gym';

```

Profile	Assume access to	Programming approach	Limitations
Full gym	Barbells, dumbbells, machines, cables, benches, pull-up station, cardio equipment	Use optimal movement choices, machines for hypertrophy, cables for isolation, precise loading	Few limitations
Dumbbells only	Fixed or adjustable dumbbells, optional bench	Use unilateral work, tempo, pauses, higher reps, mechanical drop sets	Leg loading can become difficult for advanced users
Home basic	Floor space, chair/bench, bands, light weights	Bodyweight progressions, bands, circuits, tempo, habit-building	Limited heavy loading and vertical pulling
Bodyweight only	No equipment	Push-up progressions, split squats, hinges, core, walking/running	Hard to overload legs and back long term
Bands only	Bands and optional door anchor	Rows, pulldowns, presses, RDLs, lateral raises, curls, triceps, Pallof press	Less precise loading and variable tension
Hotel gym	Inconsistent equipment, usually treadmill and dumbbells	Flexible pattern-based plans and substitutions	Equipment availability changes by location
Barbell home gym	Barbell, rack, plates, bench, maybe pull-up bar	Strength and hypertrophy basics, barbell progression	Less machine/cable isolation

7. Progression and Adaptive Programming

Templates create the first plan. Logs create the real plan. The AI should adapt based on reps, load, RPE, reps in reserve, pain, form quality, sleep, energy, and adherence.

Double progression for hypertrophy

Example:
Exercise: Dumbbell Bench Press
Sets: 3
Rep range: 8-12
Rule: When the user completes all sets at 12 reps with 1-2 reps in reserve and good form, increase load next time.

If reps = 12,12,12 and RIR $\geq$ 1: increase load.
If reps = 12,10,9: keep load.
If reps = 8,7,6: keep or reduce load.
If pain: swap exercise or reduce range/load.
If form breaks: keep or reduce load.

Strength progression model

Example:
Main lift: Squat
Top set: 1x3-5 @ RPE 8
Back-off sets: 3x5 at 90% of top-set load
Rule: If top set reaches upper rep target at RPE $\leq$ 8, add 5 lb upper body or 10 lb lower body next exposure.

If top set is easy: increase load.
If reps are missed: repeat or reduce load 2.5-5%.
If fatigue is high: reduce back-off volume.
If pain appears: switch to a safer variation.

Beginner linear progression

- Use conservative load jumps and simple targets.

- Upper-body increases: often 2.5 to 5 lb when all sets are completed.
- Lower-body increases: often 5 to 10 lb when all sets are completed.
- If dumbbells are used, increase by the smallest available jump.
- Do not chase failure. Build technique, consistency, and confidence.

Hypertrophy progression hierarchy

- First improve form, range of motion, and control.
- Then add reps within the target range.
- Then add load when the top of the rep range is achieved.
- Then add a set only if recovery is good and progress has stalled.
- Only use advanced intensity techniques for intermediate or advanced users.

8. Deloads and Periodization

```
type DeloadTrigger =  
  | 'performance_drop_2_sessions'  
  | 'persistent_soreness'  
  | 'joint_pain'  
  | 'low_sleep_3_days'  
  | 'high_stress'  
  | 'missed_workouts'  
  | 'high_fatigue'  
  | 'end_of_training_block';
```

Deload options

Deload type	How to execute	Best for
Volume deload	Reduce sets by 30-50% while keeping movement quality	Hypertrophy users, high soreness, fatigue
Intensity deload	Reduce load by 10-20%	Strength users, joint stress, nervous-system fatigue
Full recovery week	Light cardio, mobility, short full-body sessions	Overwhelmed users, poor sleep, high stress
Exercise deload	Swap aggravating lifts for lower-stress variations	Joint pain or recurring discomfort

Periodization by level

- Beginner: 8-12 weeks of foundation training with linear progression, technique, and consistency.
- Intermediate hypertrophy: 4-8 week blocks with moderate volume, progressive overload, optional peak week, and deload as needed.
- Intermediate strength: heavy/light structure or volume/intensity split with main lifts plus accessories.
- Advanced: specialization blocks, accumulation, intensification, peaking/testing, and deload or resensitization phases.

9. Nutrition and Macro Engine

The nutrition engine should not stop at calories and protein. It should produce calories, protein, carbs, fats, fiber, hydration guidance, dietary-restriction adjustments, meal timing, and adherence strategies.

```
type NutritionGoal =
  | 'fat_loss'
  | 'muscle_gain'
  | 'recomposition'
  | 'maintenance'
  | 'performance'
  | 'endurance';
```

Calorie target logic

Goal	Default calorie logic	Adjustments
Fat loss	TDEE x 0.80 to 0.90	Use conservative deficits for lean users, high stress, poor sleep, strength goals, or poor adherence
Muscle gain	TDEE x 1.05 to 1.10	Use smaller surplus for higher body fat or advanced lifters; larger for underweight or hard gainers
Recomposition	TDEE x 0.90 to 1.00	Slight deficit if higher body fat; maintenance if beginner or skinny-fat; slight surplus if lean and undermuscle
Maintenance	TDEE x 1.00	Use for performance, habit building, or diet breaks
Endurance/performance	TDEE x 1.00 or higher	Do not underfuel high-volume training; adjust carbs upward

Protein logic

Context	Target
General health	1.2-1.6 g/kg/day
Active users	1.4-2.0 g/kg/day
Fat loss	1.6-2.2 g/kg/day or 0.8-1.0 g/lb target body weight
Muscle gain	1.6-2.2 g/kg/day or 0.7-1.0 g/lb body weight
Endurance	1.2-1.8 g/kg/day
Minimum app floor	About 0.7 g/lb goal body weight unless medically constrained

Fat logic

- Use roughly 20-35% of calories from fat as a standard range.
- A practical minimum is around 0.3 g/lb body weight for many users.
- Use higher fat when the user prefers lower carb, struggles with hunger, or has lower training volume.
- Use lower fat when calories are low or the user needs higher carbs for endurance or high-volume training.
- Prioritize unsaturated fats, fatty fish, nuts, seeds, olive oil, avocado, and minimally processed sources.

Carb logic

```
carbCalories = totalCalories - proteinCalories - fatCalories
carbGrams = carbCalories / 4
```

- Carbs fill remaining calories after protein and fat.

- Higher-carb approaches generally support high-volume lifting, strength performance, sprinting, and endurance.
- Lower-carb approaches can work for fat loss if the user prefers them and performance is acceptable.
- Do not recommend keto or low carb by default for high-volume hypertrophy, strength, or endurance users.

Macro calculator pseudocode

```
function calculateMacros(user) {
  const tdee = estimateTDEE(user);
  const calories = adjustCaloriesByGoal(tdee, user.goal, user.bodyFatEstimate, user.recovery);
  const protein = getProteinTarget(user.goal, user.weightKg, user.targetWeightKg);
  const fat = getFatTarget(user.goal, user.weightLb, calories, user.preference);
  const carbs = (calories - protein * 4 - fat * 9) / 4;

  return {
    calories: Math.round(calories),
    proteinGrams: Math.round(protein),
    fatGrams: Math.round(fat),
    carbGrams: Math.round(carbs),
    fiberGrams: getFiberTarget(user)
  };
}
```

10. Dietary Restrictions and Preferences

```
type DietPreference =
  | 'no_preference'
  | 'high_protein'
  | 'vegan'
  | 'vegetarian'
  | 'pescatarian'
  | 'keto'
  | 'low_carb'
  | 'mediterranean'
  | 'halal'
  | 'kosher'
  | 'dairy_free'
  | 'gluten_free'
  | 'low_fodmap'
  | 'allergy_sensitive';
```

Diet preference logic

Preference	AI guidance
Vegan	Plan protein deliberately: tofu, tempeh, seitan, lentils, beans, edamame, soy milk, pea/soy protein. Watch B12, vitamin D, omega-3, iron, iodine, calcium, zinc, and creatine.
Vegetarian	Use eggs, Greek yogurt, cottage cheese, dairy if allowed, tofu, tempeh, legumes, protein powder. Ensure enough complete protein.
Keto/low carb	Can work for fat loss if calories and protein are controlled. Monitor performance, electrolytes, fiber, and adherence. Do not default to keto for intense training.
Gluten-free	Avoid wheat, barley, rye. Use rice, potatoes, certified gluten-free oats, quinoa, corn, fruit, legumes. Ask whether celiac or preference.
Dairy-free	Use meat, eggs, fish, tofu, tempeh, legumes, plant protein. Watch calcium and vitamin D.
Allergy-sensitive	Treat allergens as hard constraints. Do not recommend foods containing the allergen. Ask severity if meal planning.
Mediterranean	Lean proteins, fish, olive oil, legumes, vegetables, fruit, whole grains, nuts, seeds. Good default for health and adherence.

11. Fiber, Micronutrients, Food Quality, and Hydration

Fiber targets

- Women: roughly 25 g/day minimum for many adults.
- Men: roughly 30-38 g/day minimum for many adults.
- Alternative rule: about 14 g fiber per 1,000 calories.
- Increase fiber gradually to reduce digestive discomfort.
- Use fruits, vegetables, legumes, oats, whole grains, chia/flax, potatoes with skin, nuts, and seeds.

Food quality score

- Prioritize lean proteins, fruits, vegetables, whole grains, legumes, nuts, seeds, and healthy fats.
- Limit ultra-processed foods, sugary drinks, deep-fried foods, excess alcohol, and low-protein snack calories.
- For fat loss, food quality supports satiety and adherence.
- For muscle gain, food quality supports training, digestion, and health during higher calorie intake.

Hydration guidance

- Use thirst, urine color, sweat rate, climate, and training duration as practical cues.
- Increase fluids for hot climates, long cardio, high sweat rates, high-protein diets, creatine use, and low-carb diets.
- Electrolytes can be useful for heavy sweaters, long endurance, heat, keto/low-carb users, or users who feel lightheaded during training.
- Do not overprescribe water to the point of discomfort or hyponatremia risk.

12. Meal Timing and Performance Fueling

Core rule

Meal timing is secondary to total calories, protein, and adherence. It becomes more important for endurance athletes, long sessions, two-a-day training, high-volume hypertrophy, and users who underperform when training fasted.

Protein distribution

- Aim for 3-5 protein feedings per day when practical.
- A typical meal target is 25-50 g protein depending on body size and daily protein target.
- Spreading protein helps users hit targets and may support recovery better than saving most protein for one meal.

Pre-workout nutrition

- Strength/hypertrophy: protein plus carbs 1-3 hours before training works well for many users.
- Examples: Greek yogurt and fruit, protein shake and banana, chicken and rice, eggs and toast.
- Endurance: carbs matter more, especially before longer or harder sessions.
- Avoid very high fat or very high fiber right before hard training if the user gets GI issues.

Intra-workout nutrition

- Usually unnecessary for lifting under 75 minutes or easy cardio under 60 minutes.
- Useful for endurance sessions over 90 minutes, long runs/rides, sport tournaments, and multiple sessions per day.
- For prolonged endurance, 30-60 g carbs per hour is a common starting range; longer events may need more with gut training.

Post-workout nutrition

- Protein within a few hours is sufficient for most users.
- Carbs are helpful after hard training, endurance, or when training again soon.
- A shake is convenient but not mandatory if daily protein is already met.

13. Supplement Knowledge Base

Supplements are optional. The app should rank them by evidence, explain use cases, and flag cautions. Do not recommend unsafe or banned substances.

```
type SupplementEvidence =  
  | 'strong'  
  | 'moderate'  
  | 'limited'  
  | 'not_recommended'  
  | 'medical_only';
```

Supplement	Evidence	Typical use	Dose guidance	Cautions
Creatine monohydrate	Strong	Strength, muscle gain, power, repeated high-intensity work	3-5 g daily. Loading is optional.	Ask clinician for kidney disease, pregnancy, under 18, or relevant medications.
Caffeine	Strong	Performance, focus, endurance, strength, conditioning	Start 1-2 mg/kg. Common effective range 3-6 mg/kg. Usually 30-60 min pre-workout.	Anxiety, sleep issues, high blood pressure, heart conditions, pregnancy, stimulant meds.
Protein powder	Strong as convenience	Helps hit protein target	Use as needed to meet daily protein.	Allergies, digestive intolerance, dairy sensitivity.
Beta-alanine	Moderate	High-intensity efforts around 1-4 minutes, repeated sprints, combat sports, conditioning	Often 3.2-6.4 g/day split doses.	Tingling is common. Not essential for casual users.
Electrolytes	Context-dependent	Heavy sweating, heat, long endurance, keto/low-carb, lightheadedness	Use based on sweat and session length.	Avoid excessive sodium if medically restricted.
Fish oil/omega-3	Moderate	Low fish intake, general health support	Food-first or supplement if needed.	Medication interactions, bleeding risk, seafood allergy.
Vitamin D	Useful if low	Low sun exposure or confirmed deficiency	Best guided by blood test.	Avoid megadosing without clinician.
Magnesium	Useful if low	Low intake, sleep routine support, cramps if deficiency-related	Varies by form and need.	GI upset; caution with kidney disease.

Supplements to avoid recommending

- Fat burners, detox teas, SARMS, anabolic steroids, prohormones, diuretics for physique, laxatives, appetite suppressants, and unverified testosterone boosters.
- If a user asks about these, redirect to safer fundamentals: training consistency, calorie control, protein, sleep, steps, creatine, and caffeine if appropriate.

14. Coach Memory and Personalization

```
type UserMemory = {  
  goals: GoalMemory;  
  body: BodyMemory;  
  training: TrainingMemory;  
  nutrition: NutritionMemory;  
  preferences: PreferenceMemory;  
  limitations: LimitationMemory;  
  adherence: AdherenceMemory;  
  psychology: PsychologyMemory;  
  progress: ProgressMemory;  
};
```

Memory categories

Memory type	Store
Goals	Primary goal, secondary goal, desired look, timeline, motivation, event/deadline, previous failed attempts
Body	Age, sex, height, weight, waist, body-fat estimate, photos, measurements, body type notes
Training	Experience, split, equipment, days available, session length, favorite/disliked exercises, pain triggers, PRs
Nutrition	Calories, protein, diet preference, restrictions, hunger patterns, meal schedule, liked/disliked foods, weekend issues
Adherence	Missed workout reasons, best workout time, stress eating, travel, sleep issues, social drinking, motivation patterns
Progress	Weight trend, waist trend, photos, strength PRs, cardio performance, compliance score, plan changes
Psychology	Preferred coaching style, confidence level, barriers, shame triggers, accountability preference

Personalization examples

- If the user repeatedly misses Friday workouts, move the key session earlier and make Friday optional conditioning.
- If the user overeats at night, move more calories and protein toward dinner and add a planned high-protein snack.
- If strength drops too fast during fat loss, reduce the deficit slightly or reduce training volume for a week.
- If the user gets bored, change exercise variations while preserving the movement pattern and progression target.
- If the user has knee pain, remember it and stop reassigning high-stress squat/lunge variations unless cleared.

15. Workout Log Adaptation Rules

```
type WorkoutLog = {
```

```

completed: boolean;
setsCompleted: number;
repsCompleted: number[];
loadUsed: number[];
rpe?: number;
repsInReserve?: number;
pain?: boolean;
painLocation?: string;
energy: 1 | 2 | 3 | 4 | 5;
sleepQuality: 1 | 2 | 3 | 4 | 5;
notes?: string;
};

```

Session adaptation rules

Log result	Next action
All reps completed easily, good form, no pain	Increase load or reps next session according to progression model.
All reps completed but very high RPE	Keep load the same. Do not progress yet.
Missed target reps once	Repeat load next session.
Missed target reps twice	Reduce load 5-10%, reduce volume, or check sleep/calories.
Pain during movement	Stop or modify movement. Swap exercise, reduce range, reduce load, and flag limitation.
Low sleep and low energy	Reduce volume 20-30%, skip optional finishers, keep skill work easy.
Repeated missed workouts	Simplify plan, reduce days, shorten sessions, or change schedule.

Readiness score logic

```

readinessScore = average(
  sleepQuality,
  energy,
  sorenessInverse,
  stressInverse,
  motivation
)

```

8-10: normal workout, progress if performance allows.
 6-7: normal workout, moderate RPE, no extra finishers.
 4-5: reduce volume 20-30%, avoid max effort.
 1-3: recovery session, mobility, walk, or reschedule hard session.

16. Body Fat Analysis Refinements

Body-fat estimation from images should always be approximate. Use ranges, confidence levels, visual markers, and limitations. Never claim clinical accuracy.

Age-related considerations

- With age, muscle mass may decrease if untrained, fat distribution may shift toward the abdomen, and skin elasticity changes can make estimates harder.
- For older users, widen the estimate range and rely more on waist measurement, weight trend, and multiple angles.
- Avoid comparing older users directly to young physique visual guides without adjustment.

Body type adjustment

Body type	Visual clues	AI adjustment
High muscle	Large shoulders/arms/legs, visible muscle despite waist softness	Do not overestimate body fat because scale weight is high.
Low muscle / skinny-fat	Thin limbs, soft waist, low shoulder/chest/glute development	Do not recommend aggressive cutting by default. Favor recomposition.
Android distribution	More midsection storage	Weigh waist and abdomen more heavily.
Gynoid distribution	More hips, glutes, thighs	Avoid overestimating based only on lower-body fat.

Image quality rules

```

if imageQuality <= 2:
    confidence = 'low'
    rangeWidth = '8-10%'

if imageQuality >= 4 && multipleAngles:
    confidence = 'medium-high'
    rangeWidth = '3-5%'

```

Required photo guidance:

- Front relaxed
- Side relaxed
- Back relaxed
- Same lighting and distance
- Comfortable clothing
- Height, weight, age, sex
- Training history
- Waist measurement at navel

17. Program Assignment Logic

```

type ProgramInputs = {
  goal: 'fat_loss' | 'muscle_gain' | 'aesthetics' | 'strength' | 'endurance' | 'athletic';
  experience: 'beginner' | 'intermediate' | 'advanced';
  daysAvailable: number;
  equipment: EquipmentProfile;
  sessionLength: number;
  limitations: Limitation[];
  bodyFatEstimate?: string;
  recoveryScore?: number;
  preference?: string[];
};

```

Assignment rules by goal

Goal	Beginner	Intermediate	Advanced
Fat loss	3 full-body days + 2-4 cardio/steps days + simple nutrition	4 upper/lower days + 2-3 cardio days + macro precision	4-5 lifting days + planned cardio + deloads + optional diet breaks
Muscle gain	3 full-body days + double progression	4 upper/lower or 5-day hybrid + 10-20 hard sets per muscle/week	5-6 days + specialization blocks + volume cycling
Aesthetics	3 full-body days with weak-point accessories	4-day upper/lower aesthetic bias or PPL	5-day physique split with specialization

Goal	Beginner	Intermediate	Advanced
Strength	3 full-body strength days + linear progression	4-day upper/lower strength + top sets/back-offs	Periodized lift-specific blocks with peaking and deloads
Endurance	3 easy cardio days + 2 strength days	4-5 cardio days + 1-2 intensity sessions + long easy day	Polarized or pyramidal sport-specific training with fueling strategy
Athletic	Movement quality + strength basics + low-level conditioning	Strength/power/conditioning hybrid	Block-based performance plan with testing

Decision modifiers

- If days available are low, use full-body training.
- If session length is short, reduce exercise count and prioritize compound patterns.
- If recovery score is low, reduce volume and avoid high-intensity finishers.
- If equipment is limited, use unilateral work, tempo, pauses, bands, and higher reps.
- If body fat is high and goal is muscle gain, consider recomposition or a conservative fat-loss phase first.
- If the user is already very lean, avoid pushing further fat loss unless competition-specific and professionally supervised.

18. TypeScript Knowledge Module Layout

Recommended file structure

```
/src/knowledge/
  trainingKnowledge.ts
  exerciseSubstitutionKnowledge.ts
  equipmentKnowledge.ts
  progressionKnowledge.ts
  deloadKnowledge.ts
  nutritionKnowledge.ts
  macroKnowledge.ts
  dietaryRestrictionKnowledge.ts
  supplementKnowledge.ts
  coachMemoryKnowledge.ts
  bodyFatKnowledge.ts
  safetyGuardrails.ts
```

Example training principles module

```
export const trainingPrinciples = {
  universal: [
    'Specificity drives adaptation.',
    'Progressive overload is required over time.',
    'Volume must match recovery.',
    'Exercise selection should match goal, equipment, and limitations.',
    'Pain should trigger modification, not force.'
  ],
  strength: {
    load: 'Generally heavier loads, often 80%+ 1RM for primary strength work.',
    reps: '1-6 reps for primary lifts, 6-12 for accessories.',
    rest: 'Usually 2-5 minutes for heavy compounds.',
    failure: 'Avoid frequent failure on heavy compound lifts.'
  },
  hypertrophy: {
    volume: 'Use roughly 10+ hard sets per muscle group per week as a starting anchor.',
    reps: 'Most work can fall between 6-20 reps.',
    effort: 'Most sets should end around 0-3 reps in reserve.',
    frequency: 'Train each major muscle 2x per week when possible.'
```



```

    },
    fatLoss: {
      priority: 'Calorie deficit, protein, lifting, steps, cardio.',
      lifting: 'Preserve muscle with 2-4+ weekly resistance sessions.',
      cardio: 'Use sustainable cardio and daily movement. HIIT should be limited.'
    }
  };

```

Example substitution module

```

export const substitutionMap = {
  squat: {
    fullGym: ['front squat', 'leg press', 'hack squat', 'belt squat', 'goblet squat'],
    dumbbellsOnly: ['goblet squat', 'DB split squat', 'DB step-up', 'DB reverse lunge'],
    bodyweight: ['box squat', 'bodyweight squat', 'split squat', 'step-up'],
    kneeSensitive: ['box squat', 'low step-up', 'partial-range leg press', 'goblet squat to comfortable depth']
  },
  hinge: {
    fullGym: ['Romanian deadlift', 'trap bar deadlift', 'hip thrust', 'cable pull-through'],
    dumbbellsOnly: ['DB Romanian deadlift', 'single-leg DB RDL', 'DB hip thrust'],
    bodyweight: ['glute bridge', 'single-leg glute bridge', 'hamstring walkout'],
    backSensitive: ['hip thrust', 'glute bridge', 'cable pull-through', 'chest-supported back extension']
  },
  horizontalPush: {
    fullGym: ['DB bench press', 'machine chest press', 'Smith press', 'cable press'],
    dumbbellsOnly: ['DB floor press', 'DB bench press', 'DB incline press'],
    bodyweight: ['push-up', 'incline push-up', 'tempo push-up'],
    shoulderSensitive: ['neutral-grip DB press', 'machine press', 'DB floor press', 'cable press']
  }
};

```

Example supplement module

```

export const supplementKnowledge = {
  creatine: {
    evidence: 'strong',
    dose: '3-5g daily',
    timing: 'Any time of day. Consistency matters most.',
    bestFor: ['strength', 'muscle_gain', 'power', 'high_intensity_training'],
    caution: ['kidney disease', 'pregnancy', 'medication interactions', 'under 18']
  },
  caffeine: {
    evidence: 'strong',
    dose: '1-3mg/kg beginner, 3-6mg/kg experienced users',
    timing: 'Commonly 30-60 minutes pre-workout.',
    bestFor: ['strength', 'endurance', 'focus', 'conditioning'],
    caution: ['anxiety', 'sleep issues', 'heart conditions', 'high blood pressure', 'pregnancy']
  },
  proteinPowder: {
    evidence: 'strong as convenience',
    dose: 'Use to meet daily protein target.',
    timing: 'Flexible.',
    bestFor: ['fat_loss', 'muscle_gain', 'busy_users', 'low_appetite'],
    caution: ['allergies', 'digestive intolerance']
  }
};

```

19. References and Source Anchors

- American College of Sports Medicine. 2026 resistance training guideline update and resistance training prescription summaries. <https://acsm.org/resistance-training-guidelines-update-2026/>
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<https://pmc.ncbi.nlm.nih.gov/articles/PMC5596471/>
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- Dietary Guidelines for Americans 2025-2030. <https://cdn.realfood.gov/DGA.pdf>
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- ACSM physical activity guidelines resource.
<https://acsm.org/education-resources/trending-topics-resources/physical-activity-guidelines/>
- Research review on resistance training proximity to failure and adaptations. <https://pubmed.ncbi.nlm.nih.gov/38970765/>
- Review on squat depth and patellofemoral/knee stress considerations.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10987311/>
- ISSN ketogenic diet position stand.
<https://scholars.nova.edu/en/publications/international-society-of-sports-nutrition-position-stand-ketogeni/>